GATSBYVIEW RAILING BACKEND PROMPT LOGIC

Hidden prompt and decision logic to be applied automatically by GatsbyView. Consumers should only provide the simple selections defined in the UI.

Logic ID	Category	Trigger / Condition	Backend Prompt / Instruction	Consumer Input Needed	Developer / QA Guidance
B01	Core	Every railing render	Use the consumer-uploaded existing-space photo as the fixed base image. Identify the existing railing condition, open edge, stair guard, balcony edge, deck perimeter or pool-barrier condition from the photo. Modify only the railing / balustrade system and any directly required gate hardware.	None. Automatic	Foundation rule. The base photo controls architecture, geometry and scene composition.
B02	Preservation	Every railing render	Preserve the staircase, flooring, walls, windows, doors, deck or balcony structure, pool, coping, hardscape, furniture, landscaping, pets, lighting, planters and all surrounding elements exactly as shown unless the selected application specifically requires replacement of an existing guard wall.	None. Automatic	Primary background-drift control. Any unrelated scene change is a failure.
B03	Consumer Inputs	Design Your Own path	Build the hidden prompt from only four consumer selections: Application, Railing System, Handrail, and Finish. Do not require the consumer to describe placement, code, glass panel spacing, gate placement or technical installation details.	Application + System + Handrail + Finish	Keep consumer experience simple. All technical interpretation happens in backend logic.
B04	**System**	System = Base Shoe	Install clear frameless glass seated continuously in a low-profile base shoe along the required railing line. The bottom edge of each fixed glass panel must visually enter the continuous base shoe. Do not use individual spigots, floor-mounted posts or vertical framing.	Base Shoe selection	Hard system lock. If any spigots appear in fixed runs, render fails system accuracy.
B05	**System**	System = Spigot	Install clear frameless glass supported by individual floor-mounted spigots at consistent, intentional spacing along fixed glass runs. Do not add a continuous bottom channel / base shoe, posts or vertical framing.	Spigot selection	Hard system lock. Spigots should read as one consistent hardware family.
B06	**Handrail**	Handrail = Handrail	Add a true graspable handrail running along the glass at the appropriate guard height and following the railing geometry. Default to a handrail mounted along the glass rather than a cap rail sitting directly on top of the glass unless the inspiration image clearly shows a different approved treatment.	Handrail selection	Important learning from staircase testing. Do not substitute a top cap rail for a true handrail by default.
B07	**Handrail**	Handrail = No Handrail	Do not add a handrail, cap rail or top rail. Keep the top edge of the glass visually clean unless gate hardware or another required connection is present.	No Handrail selection	Hard negative.
B08	**Finish**	Every render	Apply the consumer-selected finish consistently to all visible railing hardware, including base shoe or spigots, handrail, clamps, connectors and gate hardware.	Finish selection	Explicit consumer finish overrides the finish shown in an inspiration image.
B09	Existing Railing	Existing railing / guard is visible	Remove only the existing railing system and replace it along the same railing location, stair slope, landing geometry, balcony edge or deck perimeter. Preserve the supporting architecture and do not extend the new railing onto adjacent walls or surfaces.	None. Automatic from image	Prevents railing from being added where drywall or architecture should remain.
B10	Open Edge	No existing railing at a clear open edge	Add the selected glass railing only along the existing open edge or guard line. Do not reshape, relocate or extend the architectural edge.	None. Automatic from image	Use existing geometry as placement anchor.
B11	Interior Balcony + Stair	Application = Interior balcony and staircase	Evaluate the full visible guard condition. If both a balcony / landing edge and stair flight require railing, apply the selected system to both. Follow the existing stair slope and landing geometry. Do not omit the stair railing and do not create railing over intact drywall or full-height walls.	Application selection	Addresses prior failure where balcony was changed but stair flight was missed, then a later render replaced drywall incorrectly.
B12	Interior Stair / Half Wall	Application = Interior staircase AND a guard-height half wall / pony wall clearly occupies the stair guard location	When the existing half wall is functioning as the stair guard, remove only that guard-height wall along the stair run and replace it with the selected glass railing. Preserve full-height walls, adjacent structure, trim, flooring, ceiling and stair geometry.	Application selection. Automatic image interpretation	Use only when the wall is clearly the existing guard condition. Do not remove unrelated drywall.
B13	Exterior Deck / Patio	Application = Exterior deck / patio	Use the existing deck or patio perimeter and existing railing line as the guide. Replace only the railing. Preserve decking, stairs, furniture, house, landscaping, pets, lighting, planters and surrounding structures exactly as shown.	Application selection	If no railing is present, follow the clearly defined open deck / patio edge rather than inventing a new perimeter.
B14	**Pool . Existing Fence**	Application = Pool balustrade / pool fence AND an existing fence is visible	Use the existing pool-fence line as the placement guide and replace that barrier with the selected glass system. Preserve the pool, coping, deck, landscaping, furniture and architecture. Retain a functional gate at the existing or most logical access location.	Application selection. Automatic image interpretation	Existing fence provides the placement anchor. Do not move the enclosure closer to the pool.
B15	**Pool . No Existing Fence**	Application = Pool balustrade / pool fence AND no existing barrier is visible	Create a code-conscious glass pool enclosure on the surrounding usable deck / patio area. Do not trace or snake around the exact shape of the pool. Favor clean, straighter runs and intentional corners, positioned back from the pool edge to preserve practical circulation space. Do not place the barrier directly on the pool coping unless site geometry clearly requires it.	Application selection. Automatic image interpretation	Dedicated pool-placement logic. Consumer should never be asked to determine placement. Final engineering / code compliance must be verified locally.
B16	**Pool Gate**	Every pool balustrade render	Include at least one clearly identifiable, functional access gate at a logical pedestrian approach to the pool area. Use appropriate hinges and latch hardware. The gate must read visually as operable and distinct from fixed glass panels.	None. Automatic	Gate is mandatory background logic for pool visualizations, even when consumer does not mention it.
B17	**Pool Gate . Base Shoe**	Pool + Base Shoe	Run the continuous base shoe only beneath fixed glass panels. Interrupt the base shoe at the operable gate opening. Do not place a base shoe, spigot or other fixed support directly beneath the gate panel.	None beyond Base Shoe selection	Critical gate / system interaction.
B18	**Pool Gate . Spigot**	Pool + Spigot	Use spigots only beneath fixed glass panels. Do not place a spigot directly beneath the operable gate panel. Use gate-specific hinges / latch hardware at the gate.	None beyond Spigot selection	Critical gate / system interaction.
B19	**Pool Placement Guardrail**	Pool with no existing fence	Use the property hardscape and circulation pattern to infer a practical enclosure zone. Keep the pool, pool steps and water completely unobstructed. Favor a perimeter that reads as a fence around the pool area, not a railing attached to the pool itself.	None. Automatic	QA should specifically flag “too close to pool edge” and “snaking with pool shape.”
B20	**Match Inspiration**	Consumer Path = Match Inspiration	Use the consumer existing-space photo as the base image and the uploaded inspiration image only as the visual reference for railing system appearance, glass configuration, handrail treatment, hardware style and gate treatment where applicable. Preserve the base-image architecture and surroundings.	Existing-space photo + inspiration image	Inspiration controls STYLE. Base image controls SPACE / GEOMETRY.
B21	**Match Inspiration . Placement**	Match Inspiration + any application	Do not copy the railing path, dimensions, background or architecture from the inspiration image. Determine railing placement from the base image and the application-specific placement logic.	None. Automatic	Especially important for pools. Inspiration should never dictate where the fence sits in the customer property.
B22	**Match Inspiration . Overrides**	Match Inspiration + explicit consumer option override	When the consumer explicitly selects a finish or other available option that differs from the inspiration image, use the explicit consumer selection. Use the inspiration image for all remaining style attributes.	Any explicit override selection	Precedence: explicit consumer selection > inspiration image > backend defaults.
B23	Design Your Own	Consumer Path = Design Your Own	Do not require, request or assume an inspiration image. Generate from the existing-space photo plus Application, System, Handrail and Finish selections using the approved GatsbyView system logic.	Four standard selections	Reference image = none is valid and expected for this path.
B24	Refinement	Consumer requests a revision after first render	Use the most recent rendered image as the new fixed base. Lock all elements the consumer says are correct. Make only the specifically requested change and preserve every other element exactly as shown.	Revision request only	One-change-at-a-time refinement minimizes drift.
B25	Background Drift	Every render	Do not move, remove, add, recolor or redesign unrelated objects or surfaces. Do not alter camera viewpoint, crop, architecture, landscaping, furniture, pool shape, hardscape, pets or lighting simply to accommodate the railing.	None. Automatic	Any unintended scene change should be logged as background drift.
B26	Panel / Hardware Rhythm	Every render	Keep glass panels, joints, spigots, clamps and base-shoe runs visually consistent, evenly organized and realistic for the available geometry. Avoid random support spacing, unnecessary panel breaks or mixed hardware systems.	None. Automatic	Conceptual visual consistency only. Exact engineering dimensions are not inferred from a photo.
B27	Approved System Vocabulary	Every render	Restrict railing visuals to Gatsby Glass approved system families: Base Shoe or Spigot, with approved handrail treatment and selected finish. Do not invent cable rail, picket rail, framed-post systems or unrelated railing types unless the inspiration path explicitly maps to an approved Gatsby configuration.	None. Automatic	Approved hardware vendors / visual references: CRL, FHC, Q-railing.

B28	Vendor Reference	Developer reference	Use approved vendor systems as the visual / technical reference library when creating system-specific examples and prompt conditioning: CR Laurence, FHC, and Q-railing.	None	https://www.crlaurence.com/ | https://fhc-usa.com/ | https://q-railing.com/en-us/
B29	Code / Safety Disclaimer	Every render	Treat the output as a design visualization, not an engineered or code-approved installation drawing. Backend logic should be code-conscious, but final railing height, glass specification, attachment method, gate requirements, setbacks and local code compliance must be confirmed by the local Gatsby Glass team.	None. Automatic	Important because code requirements vary by jurisdiction and cannot be confirmed from a photo alone.
B30	Prompt Assembly Order	Developer implementation	Assemble the hidden prompt in this order: fixed-base + preservation rules; application-specific placement logic; selected system rule; handrail rule; finish rule; pool gate rule if applicable; inspiration logic if applicable; final no-drift guardrail. Keep wording concise and avoid duplicating the same instruction repeatedly.	None	The consumer should never see or type this assembled backend prompt.

IMPLEMENTATION PRINCIPLE: Keep consumer-facing inputs simple. All placement, system interpretation, gate logic, preservation rules and application-specific decisions belong in the hidden GatsbyView prompt assembly layer.